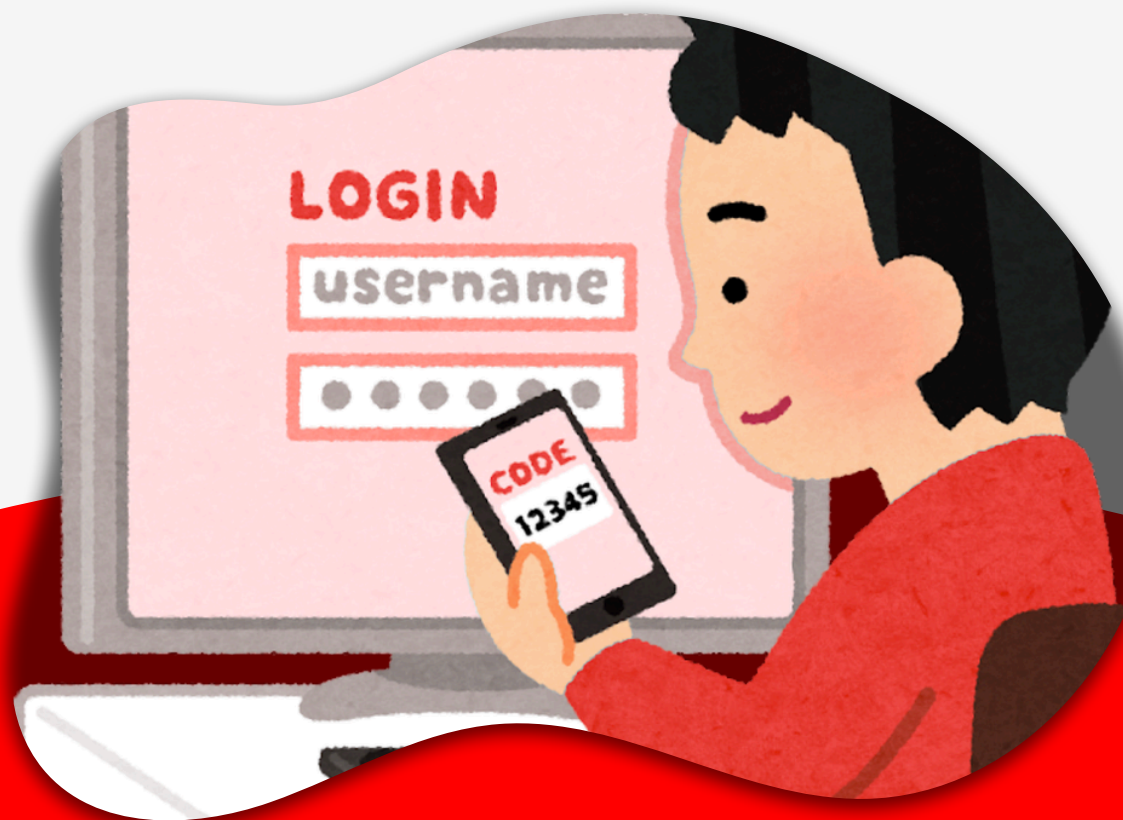


MULTI-FACTOR AUTHENTICATION

PROTECTING YOUR IOWA STATE UNIVERSITY

ACCOUNT

USER MANUAL



AUTHOR: MUBINA SADRIDINOVA

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INSTITUTION: IOWA STATE UNIVERSITY

031 Parks Library
701 Morrill Rd
Ames, IA 50011



INTRODUCTION

This guide is for Iowa State University students, faculty, and staff who use systems such as Outlook, Microsoft 365, and Canvas. It explains how Multi-Factor Authentication (MFA) works and why it is required for account security. This document describes how MFA protects university accounts and how users complete authentication during login. It covers the structure and function of MFA but does not include administrator setup.

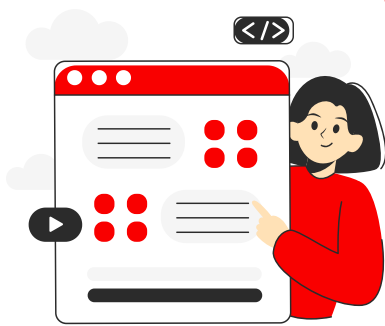
Multi-Factor Authentication (MFA) verifies identity using two or more independent factors, usually a password and confirmation from a registered device. Its purpose is to prevent unauthorized access even if a password is compromised. MFA was introduced to strengthen security as password-only systems became vulnerable to cyberattacks. MFA is required at Iowa State University to protect academic, email, and institutional systems.



DEFINITION

Multi-Factor Authentication is a layered identity verification system requiring users to provide two or more independent authentication factors. The system evaluates each factor separately, and access is granted only when all required factors are successfully verified.

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Comparison and Contrast

Comparison and Contrast

Single-Factor Authentication

Uses only password

Lower security

Easier to compromise

Multi-Factor Authentication

Uses password + second factor

Higher security

Much harder to compromise

Analogy

MFA works like a door with two locks. Even if someone unlocks the first lock (password), they still need the second lock (phone verification) to enter.

Negation

MFA is not:

- Only password authentication
- Only biometric verification
- Guaranteed absolute security
- It is a layered security system that reduces risk.

Etymology

- Multi = more than one
- Factor = independent verification method
- Authentication = identity verification



Purpose of the Mechanism

The purpose of Multi-Factor Authentication (MFA) is to **improve system security** by preventing unauthorized access. Password-only authentication is vulnerable to phishing, brute-force attacks, and credential theft. MFA adds an additional verification layer, ensuring that access is granted only when multiple identity factors are verified. MFA ensures that even if a password is compromised, unauthorized users cannot access protected systems without completing the second authentication factor.

MFA protects:



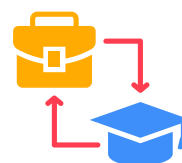
- University accounts



- Email and cloud systems



- Academic records and personal data



- Institutional resources



Something You Know:

Step1. Enter User's Net-ID

Step2. Enter User's Password



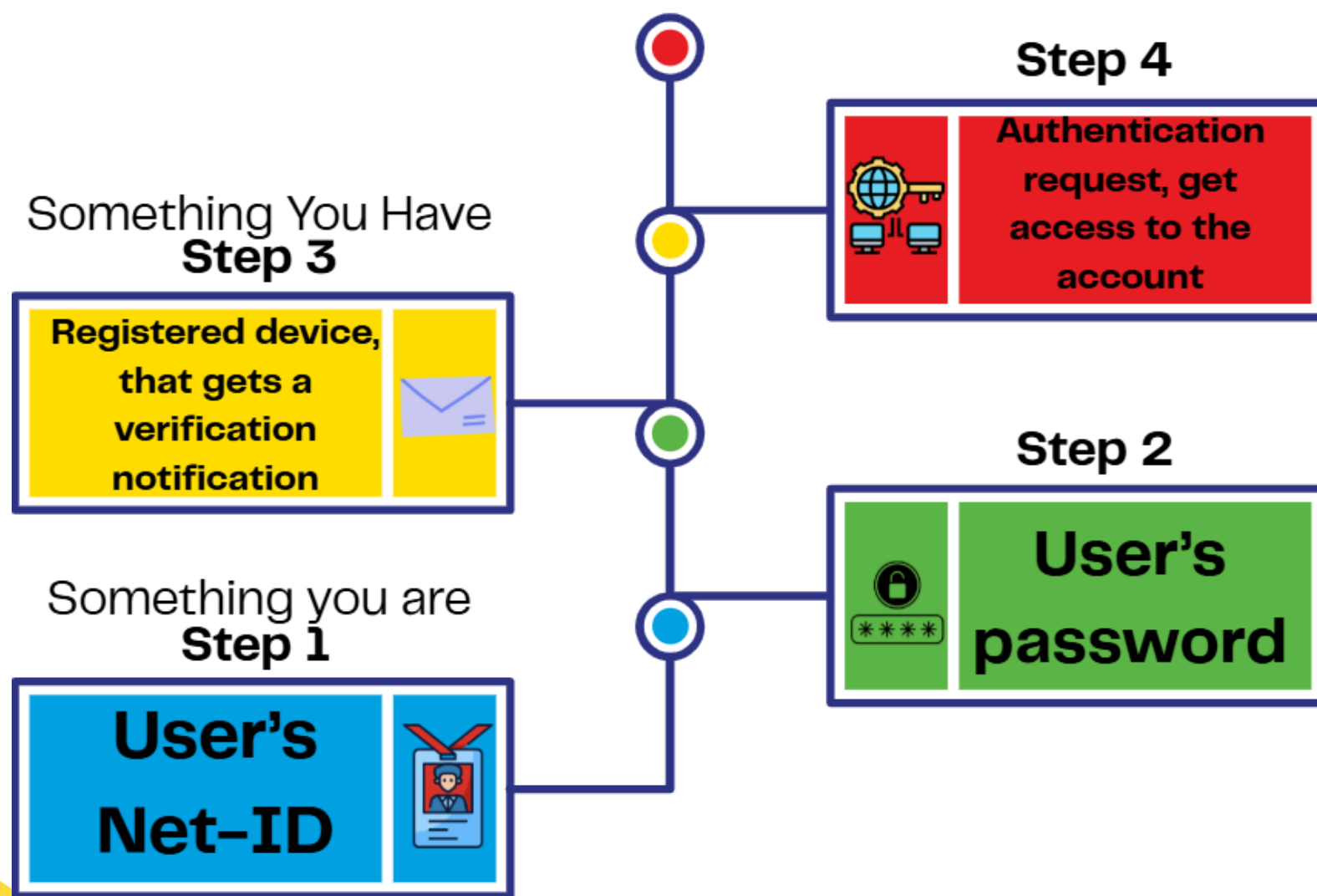
WARNING

Never approve a Multi-Factor Authentication request that you did not initiate. **Unexpected verification prompts may indicate unauthorized access attempts.** Approving such requests can allow unauthorized access to your account and personal data.

Something You Have:

Step3. Registered device, such as smartphone or laptop, that gets verification notification and code.

Step4. Authentication request, get access to the account.



Multifactor Authentication Methods and How to Enroll

CHOOSE THE MFA METHOD RIGHT FOR YOU



Iowa State University requires two forms of multifactor authentication (MFA) to keep your account secure. **MFA acts as an additional layer of security to prevent unauthorized users from accessing your account** when your password has been compromised. Below are the MFA options available to Iowa State University users.

Microsoft Authenticator Mobile App

Available on Android and iOS, Microsoft Authenticator can send you push notifications to confirm account logins, or you can use the app to obtain a temporary 6-digit code to approve an account login.



Microsoft Authenticator Mobile App

1. **Sign in to** <https://mysignins.microsoft.com/security-info>, with **Iowa State email and password**. You will be prompted with a page similar like in figure 1.

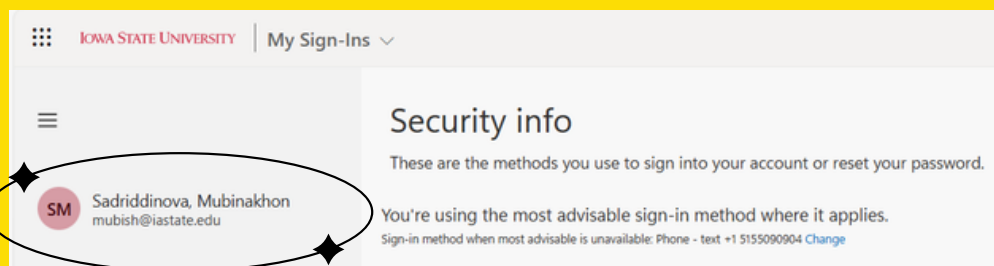


Figure 1.

2. Click **Add sign-in method**.

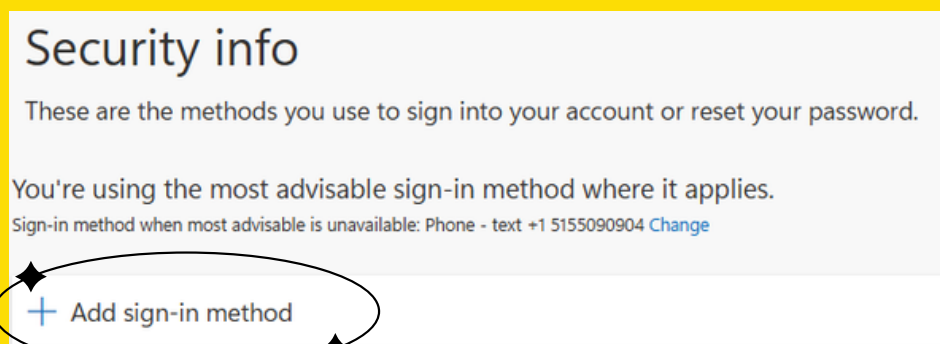


Figure 2.

3. Choose **Microsoft Authenticator**.

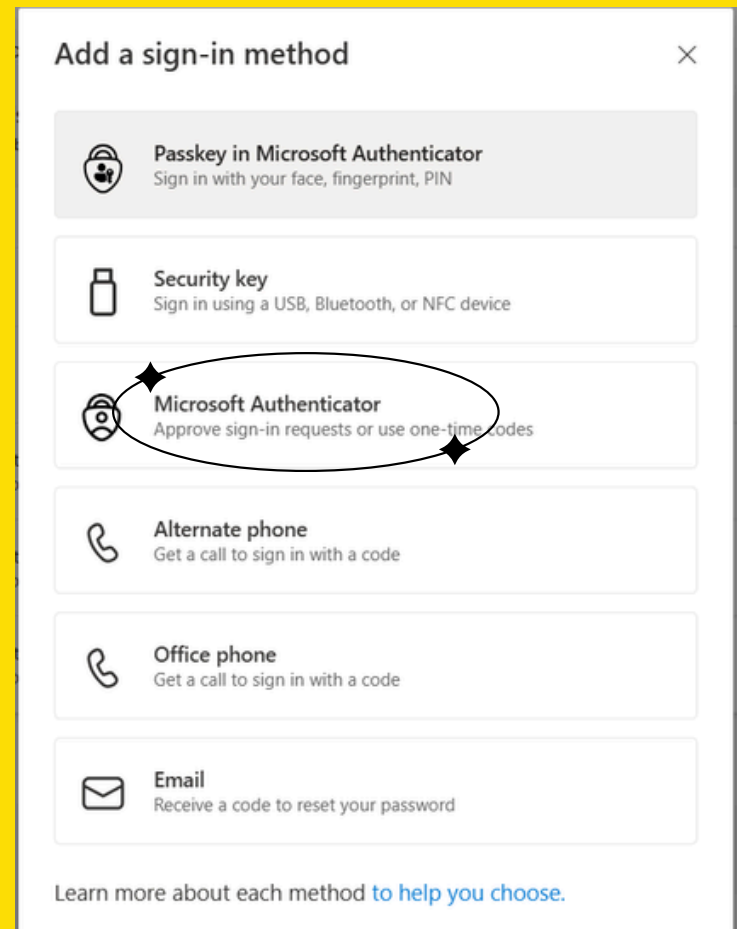


Figure 3.

The next steps are in the following pages

NOTE

Complete the steps above on your laptop or desktop to initiate the enrollment process.

- Microsoft Authenticator Mobile App
- Google Authenticator Mobile App
- Phone
- Security Key



4. **Download the Microsoft Authenticator App on your phone.** You can find that going to the “**App Store**” or “**Google Apps**”, depending on the phone you use. **In the search bar write “Microsoft Authenticator”** as shown in the figure 4, then **click on the Microsoft Authenticator. Click “Get” or “Download”** as in the figure 5.

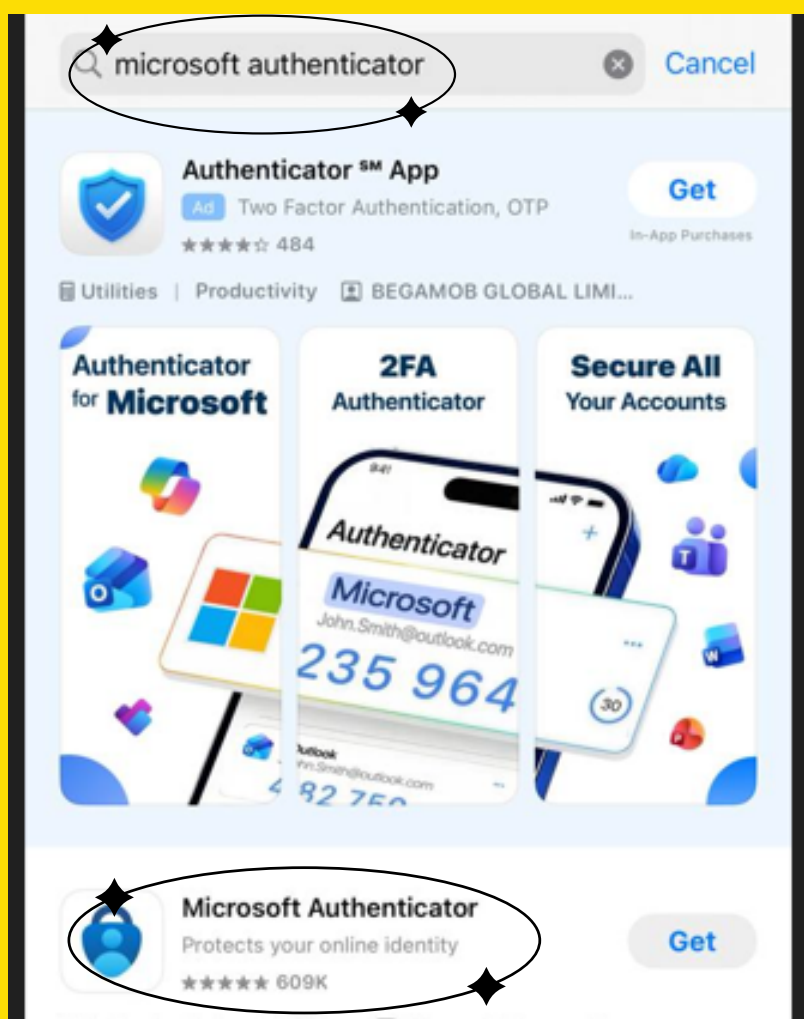


Figure 4.

NOTE

All of the examples were done **on iPhone**; your screen might look different.

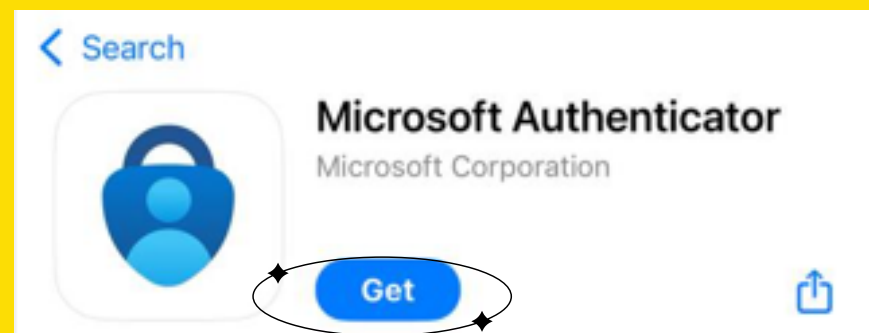


Figure 5.

After you download app, you can move on to the next step

5. Choose **Next**.

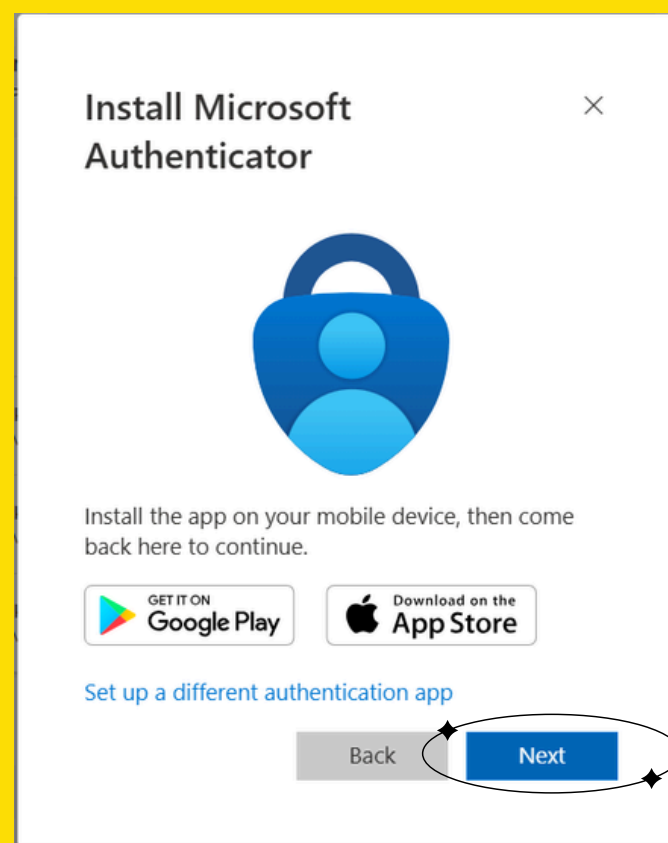


Figure 6.



6. Choose **Next**.

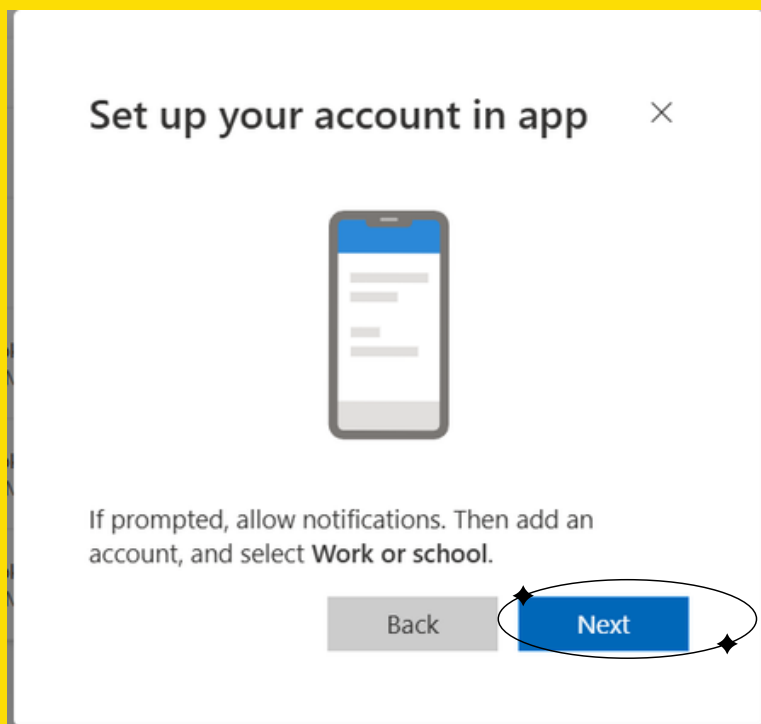


Figure 7.

7. Open the **Microsoft Authenticator app** on your phone and **click the + button at the top**. Then **choose Work or School Account**.

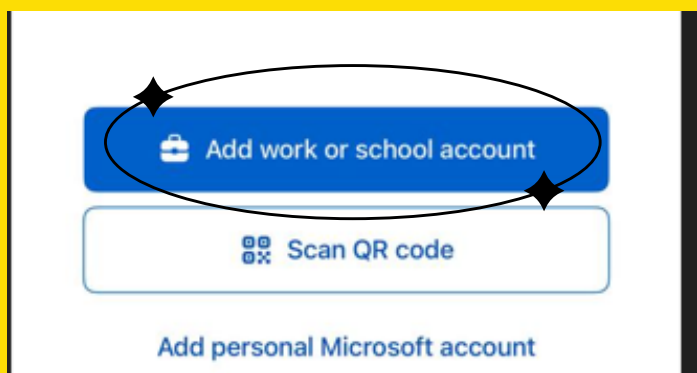


Figure 8.

8. Choose scan a QR code in the App and **scan the code that is displayed** OR Login. Then click **Next**.

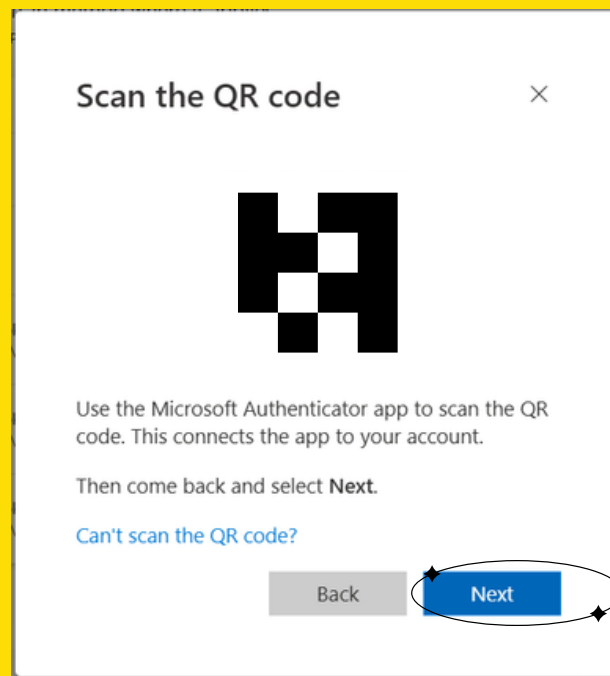


Figure 9.

9. After you are done, **Microsoft Authenticator will show in the list on your Security Info page**.

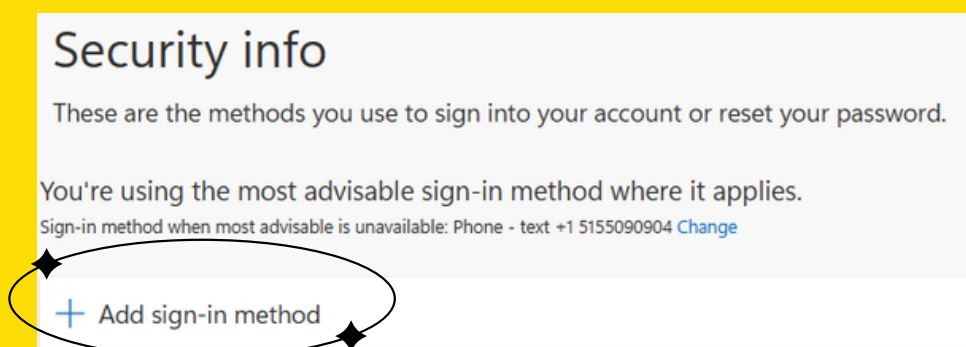
NOTE

For **Security reasons** not an actual QR code was placed and **some steps**, has **not been shown visually**, but explained in the best way possible

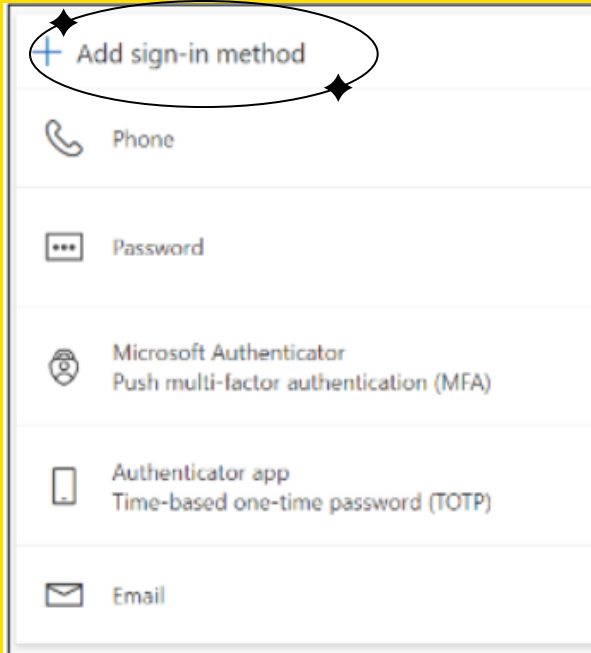


Phone Call

1. Go to <https://mysignins.microsoft.com/security-info>.
2. Enter **your ISU Email address** if prompted
3. Once on the Security info page, click '**Add sign-on method**'.



4. On the **Add a method page**, select **Phone**, and then select **Add**.



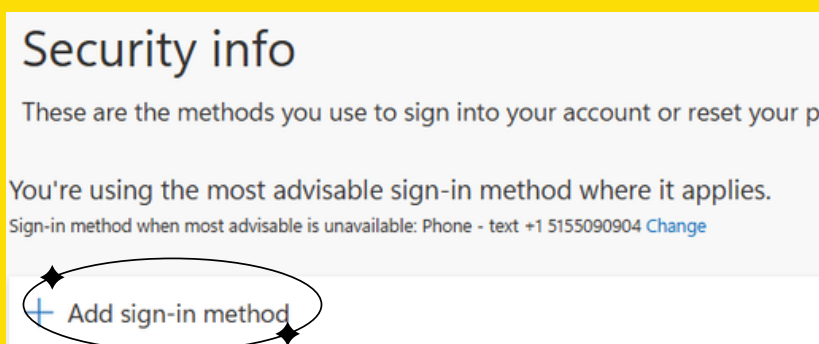
5. On the Phone page, **type the phone number for your mobile device**, choose **Call me**, and then select **Next**.
6. **Answer the verification phone call**, sent to the phone number you entered, and **follow the instructions**, prompted afterwards.



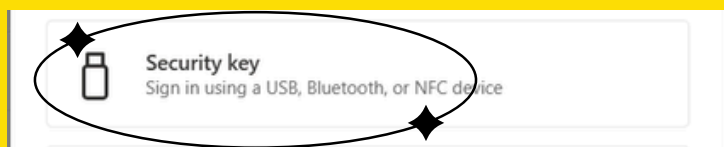
Security Key

1. Go to <https://mysignins.microsoft.com/security-info>.

2. Click **+Add sign-in method**.



3. Choose **'Security Key'** in the pop-up menu.



4. For security purposes, you will be asked to sign in with two-factor authentication. **Choose 'Next'**. Confirm your MFA prompt. NOTE: **If you don't have any factors set-up, you will not get this prompt.**

5. Choose the type of **security key (YubiKey)** you are configuring. When prompted, prepare your YubiKey and **insert it into your device**. Select **Use an external security key** when asked where to save your passkey.

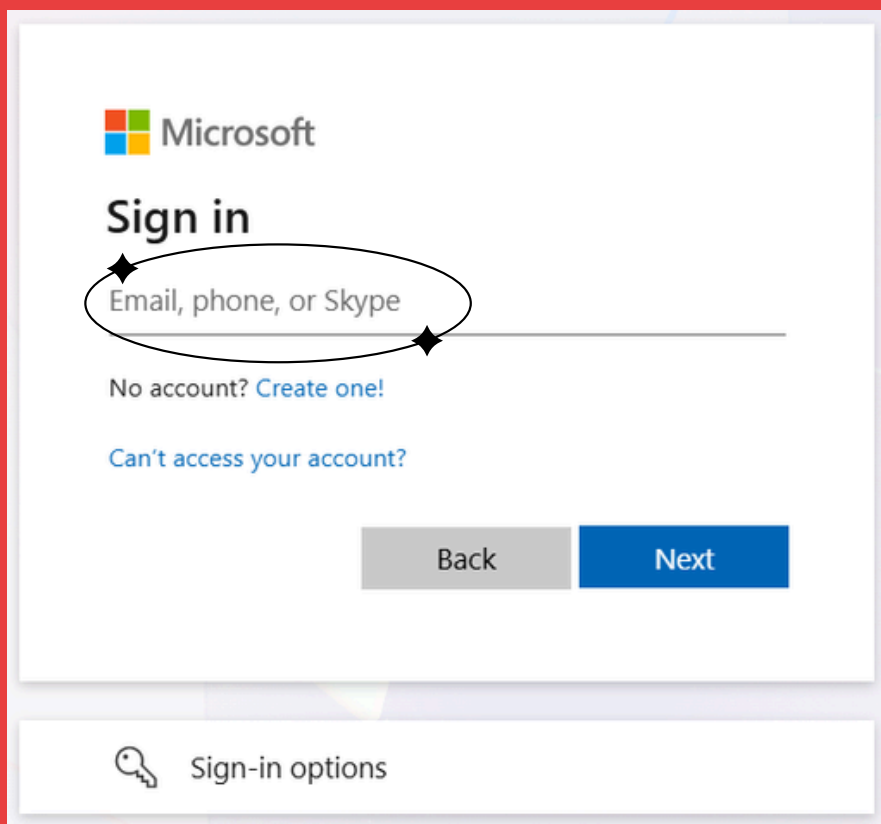
6. **Confirm setup** to allow your YubiKey to sign in to your Iowa State account. If prompted, **create or enter your YubiKey PIN**. Finally, **name your device using a simple and recognizable label**.



Principles of Operation

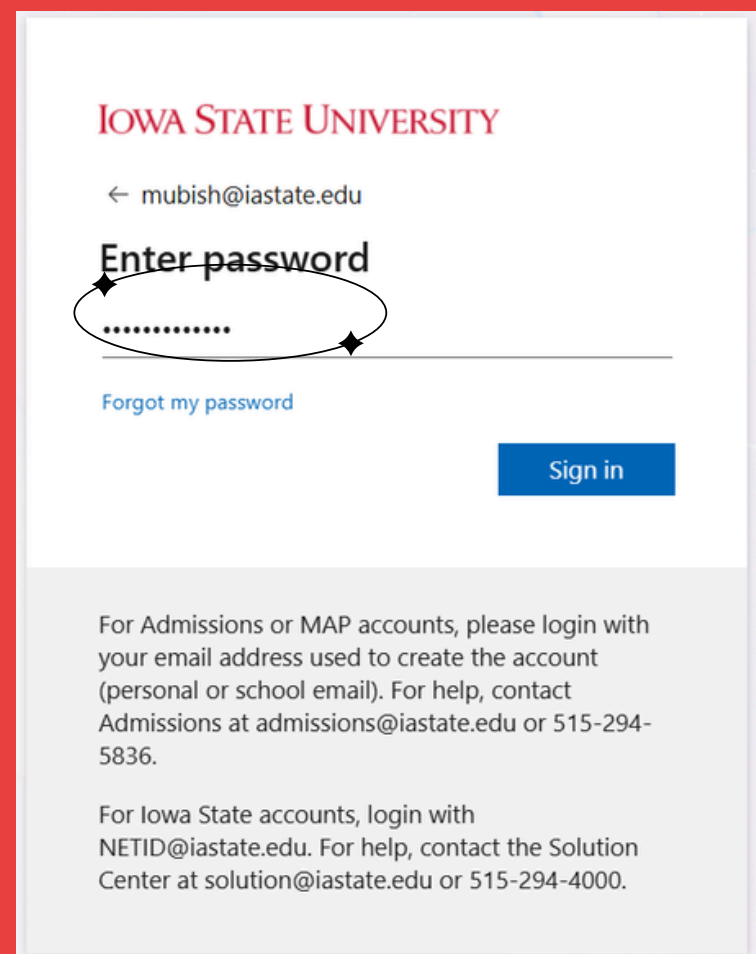
The MFA system operates in the following steps:

1. The user enters their **Net-ID** and password.



1.1 Entering **ISU Net-ID**

When logging in, enter your ISU Net-ID in the field labeled email, phone, or Skype.



1.2 Entering Password

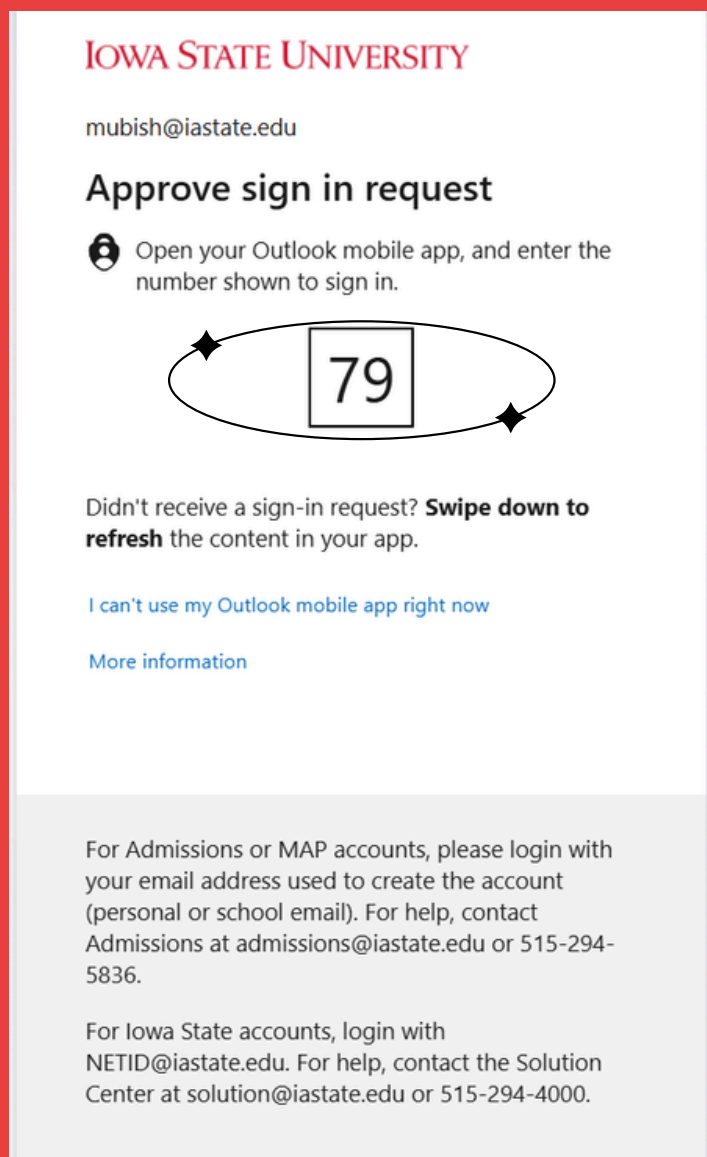
Enter the password associated with your ISU account..

NOTE

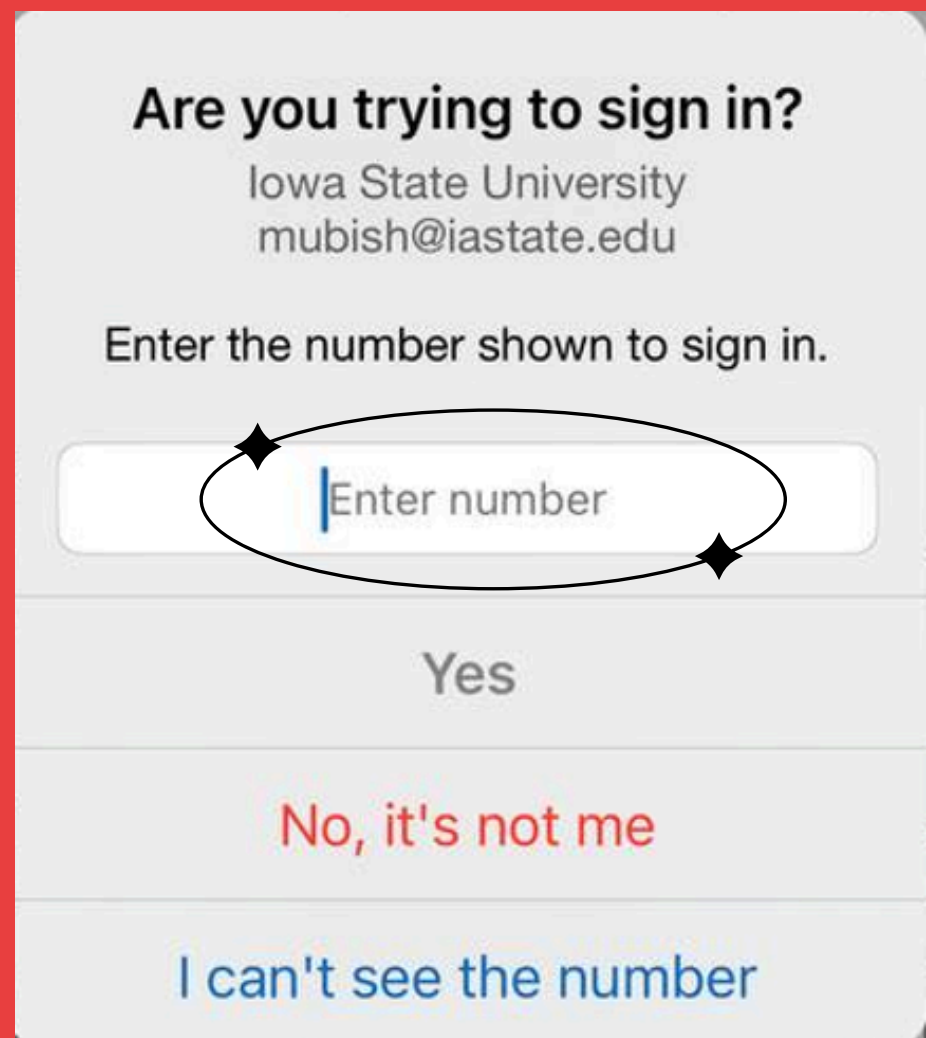
After entering the password, the system verifies your credentials before proceeding to the next step.



2. The system verifies the password and sends a second authentication request.



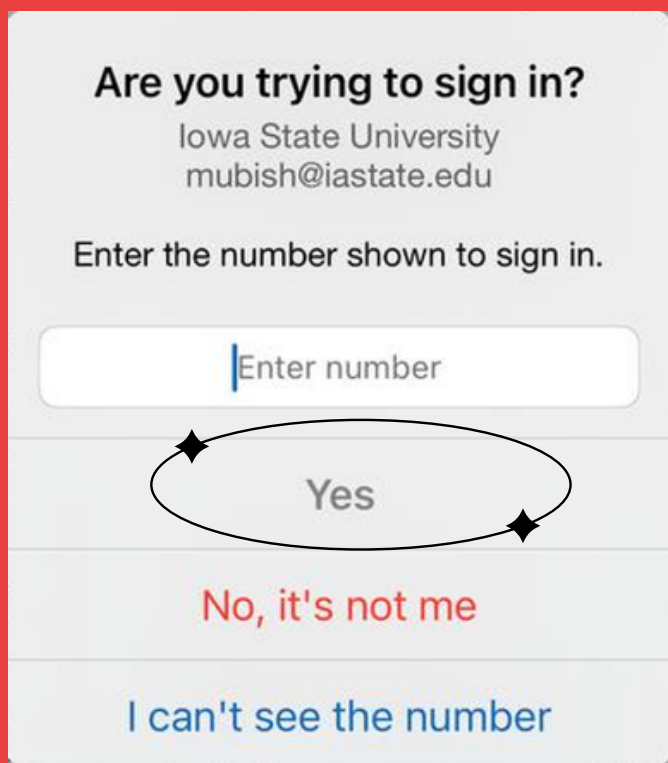
2.1 The system **sends a verification prompt** to the Outlook or Authenticator app.



2.2 **Enter the displayed number** in your app to continue



3. The **user confirms the request** using a registered MFA method. **Access is granted** only after successful verification.



Are you trying to sign in?
Iowa State University
mubish@iastate.edu

Enter the number shown to sign in.

Enter number

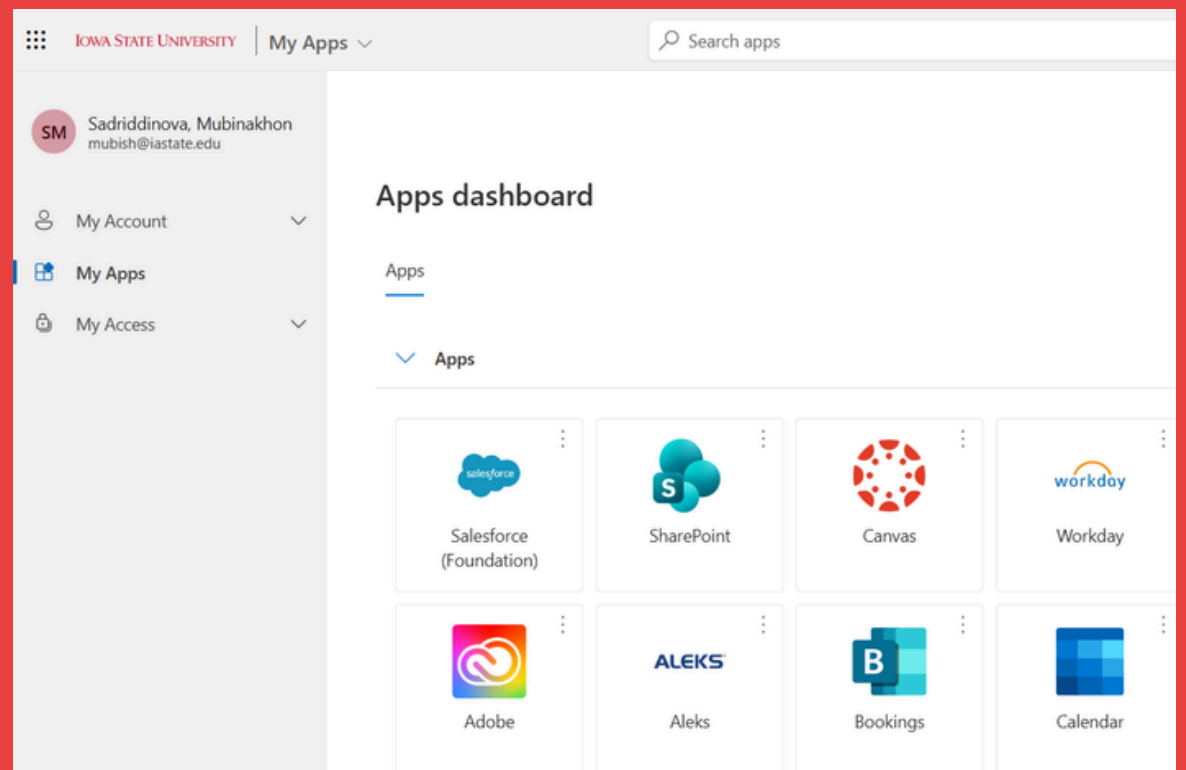
Yes

No, it's not me

I can't see the number

The image shows a mobile app interface for MFA verification. It asks the user if they are trying to sign in for Iowa State University. After entering a number, there are three buttons: 'Yes' (circled with a black oval and arrows), 'No, it's not me', and 'I can't see the number'.

3.1 After entering the number, **select Yes** to confirm the login request.



3.2 You will then be **granted access** to your My Apps dashboard



WARNING

Do not select “**No, it's not me**” if you initiated the login request.



User Instructions

Follow these steps:

1. Go to the **Iowa State login page**.
2. Enter your **Net-ID and password**.
3. Wait for the **MFA verification** request.
4. **Approve the request** using your configured authentication method.
5. **Access your** account after successful verification.
6. Only **approve requests** you personally initiated.

Conclusion

Multi-Factor Authentication provides a **strong layered security system that protects user accounts and sensitive university data**. By **combining password verification with device authentication**, MFA significantly reduces the risk of unauthorized access. Maintaining updated authentication methods and approving only legitimate login requests ensures continued account security. MFA remains an essential protection mechanism for modern digital systems.



WARNING

Security Notice:

Never approve an MFA request that you did not initiate. Unexpected verification prompts may indicate unauthorized access attempts.



Troubleshooting

1.No verification received

- Check your internet connection
- Ensure your device is powered on.

2.Lost or replaced phone

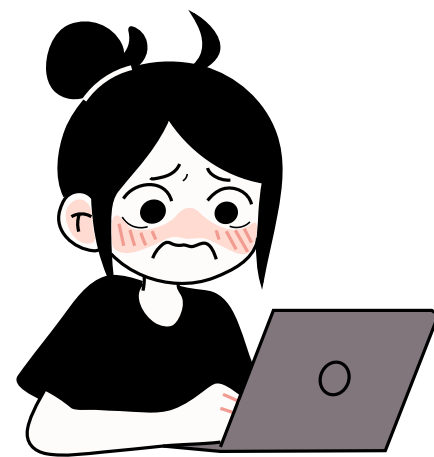
- Use a backup authentication method
- Contact IT support to re-register your device.

3.Unexpected MFA request

- Do not approve.
- Change password immediately.

4.Unable to log in

- Verify your authentication method is configured correctly
- Try again.



WARNING

If you receive an **MFA request without logging in**, deny the request and **change your password immediately**.



REFERENCES

MLA Citation

MFA Verification Methods – Information Technology Services.
www.it.iastate.edu/mfa-methods.

What Is: Multifactor Authentication – Microsoft Support.
support.microsoft.com/en-us/topic/what-is-multifactor-authentication-e5e39437-121c-be60-d123-eda06bddf661.

Information Technology Services — IT Solution Center

031 Parks Library
701 Morrill Rd
Ames, IA 50011

515-294-4000
solution@iastate.edu

